

CFD-HW3

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Outline

In this assignment, I was tasked with meshing the interior of a pipe with a sudden diameter contraction along its length. From there, CFD analysis in Fluent would be applied in order to model the fluid dynamics inside the piping. It is assumed that incompressible water enters from one end with an initial constant velocity at the inlet boundary with a Reynold's number of 1700.

To be specific, I was tasked with simulating the flow pattern along the pipe via stream lines, developing a pressure field, and observing the (planar) velocity profile 1 inch away from the contraction given three meshing criteria as will be laid out later.

Geometry

The provided geometry file depicted a hollow pipe formed by two connected cylinders. The first cylinder has a diameter of $D_1 = 111.66$ mm and a length of $L_1 = 609.6$ mm. The second (smaller) cylinder has a diameter of $D_2 = 49.63$ mm and a length of $L_2 = 307.98$ mm.

In any case, the enclosed volume needed to be meshed, not the pipe itself. So, I had to go into the geometry (SpaceClaim) and extract the volume by selecting the respective surface boundaries. Instead of deleting the pipe (shell), I decided that I'd just suppress it in the meshing tool for convenience.

From there, I exited the geometry editor, and was almost ready to begin meshing. In the meshing tool, I carefully named the important boundary faces so I could later apply the boundary conditions.

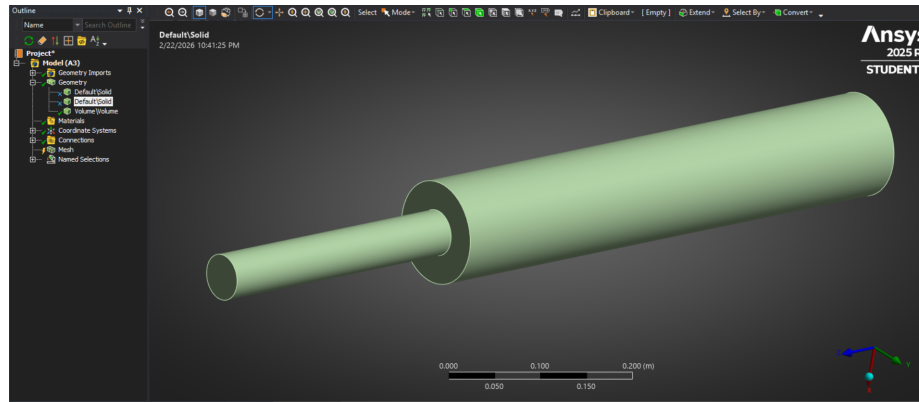


Figure 1: Preliminary Geometry Work (symmetry split)

Preliminary Calculations

Before performing the meshing and CFD analysis, a baseline calculation ought to be performed. To start, water has a density $\rho = 1000 \text{ kg/m}^3$ and viscosity $\mu = 0.001 \text{ Pa} \cdot \text{s}$. Because the initial Reynolds number is $\text{Re}_1 = 1700 < 2300$, the flow is laminar. The respective friction factor is roughly $f_1 \approx 0.038$ (Moody diagram). The inlet speed at the inlet can be calculated as such:

$$V_1 = \frac{\mu \text{Re}_1}{\rho D_1} = \frac{(0.001 \text{ Pa} \cdot \text{s})(1700)}{(1000 \text{ kg/m}^3)(0.11166 \text{ m})} \approx 0.015 \text{ m/s}$$

By conservation of mass, $\rho V_1 A_1 = \rho V_2 A_2$, where A_1 and A_2 are the inlet area and contraction interface area. Therefore, it follows that:

$$V_2 = \frac{A_1}{A_2} V_1 = \left(\frac{D_1}{D_2} \right)^2 V_1 \approx \left(\frac{111.66}{49.63} \right)^2 (0.015 \text{ m/s}) \approx 0.077 \text{ m/s}$$

The small pipe is expected to have the following Reynolds number as given below:

$$\text{Re}_2 = \frac{\rho V_2 D_2}{\mu} \approx \frac{(1000 \text{ kg/m}^3)(0.077 \text{ m/s})(0.04963 \text{ m})}{0.001 \text{ Pa} \cdot \text{s}} \approx 3820 \implies f_2 \approx 0.039 \implies f_1 \approx f_2 \approx f = 0.04$$

The net minor pressure drop (due to friction) can now be computed as follows:

$$\Delta P_{\text{minor}} = f_1 \frac{L_1}{D_1} \frac{\rho V_1^2}{2} + f_2 \frac{L_2}{D_2} \frac{\rho V_2^2}{2} \approx \frac{(0.04)(1000 \text{ kg/m}^3)}{2} \left(\frac{609.6}{111.66} (0.015 \text{ m/s})^2 + \frac{307.98}{49.63} (0.077 \text{ m/s})^2 \right) \approx 0.74 \text{ Pa}$$

The net major pressure drop (due to the sudden change in geometry), and thus the overall net pressure drop, can now be computed as such (with $K_c \approx 0.4$):

$$\Delta P_{\text{major}} = K_c \frac{\rho V_2^2}{2} \approx (0.4) \frac{(1000 \text{ kg/m}^3)(0.077 \text{ m/s})^2}{2} \approx 1.2 \text{ Pa} \implies \Delta P_{\text{net}} = \Delta P_{\text{minor}} + \Delta P_{\text{major}} \approx 2 \text{ Pa}$$

The First Mesh

The first mesh demanded a nominal mesh element size of $0.2'' = 0.00508 \text{ m}$, and that it be unstructured. By default, this entailed using the sweeping tool in Fluent.

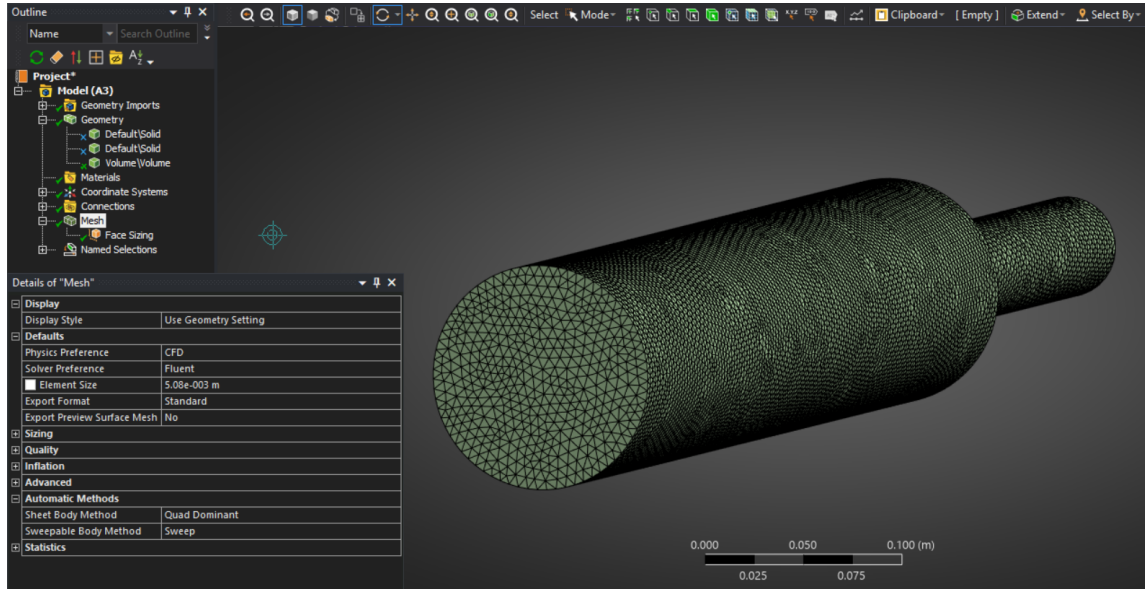


Figure 2: The First Mesh

Upon implementing the mesh into Fluent, the respective fluid (water) was assigned to the cell zones of interest and the boundary conditions were executed. To be precise, the velocity was initialized at the inlet (0.015 m/s), and the turbulence settings were fixed at the inlet/outlet. The k-omega (SST) model was engaged, and the model was ready for initialization. Below is a residual plot, which demonstrates convergence.

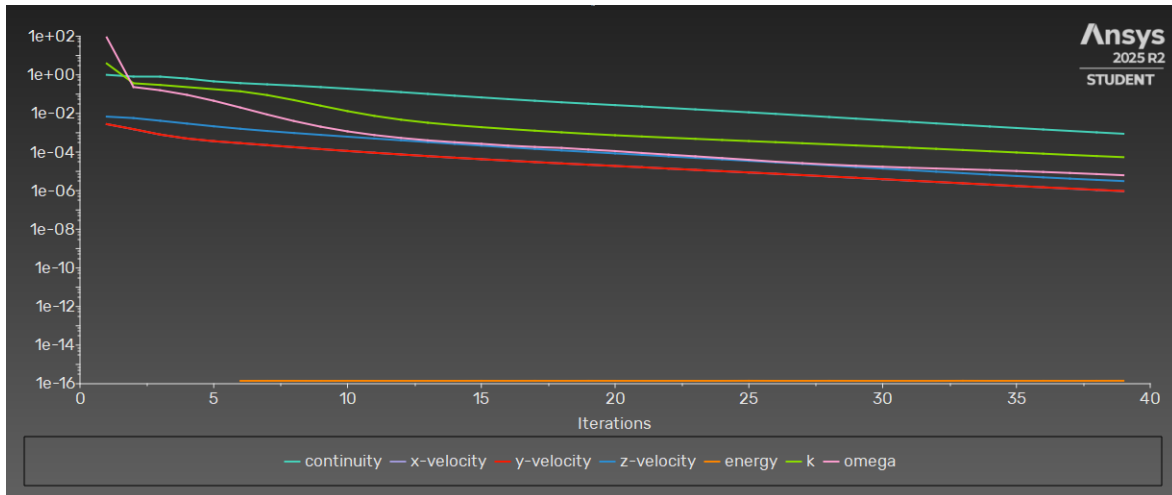


Figure 3: Residuals with Respect to the First Mesh

The streamlines in the pipe (colored by fluid speed) were determined to be as such:

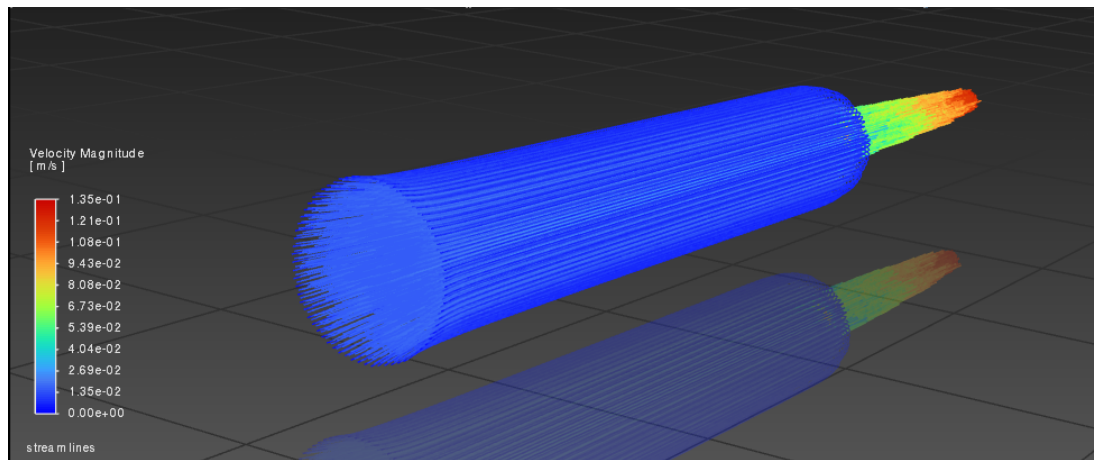


Figure 4: Stream Lines with Respect to the First Mesh

The speeds do appear to match the predicted values relative to their orders of magnitude. The pressure contour over the entire pipe follows below:

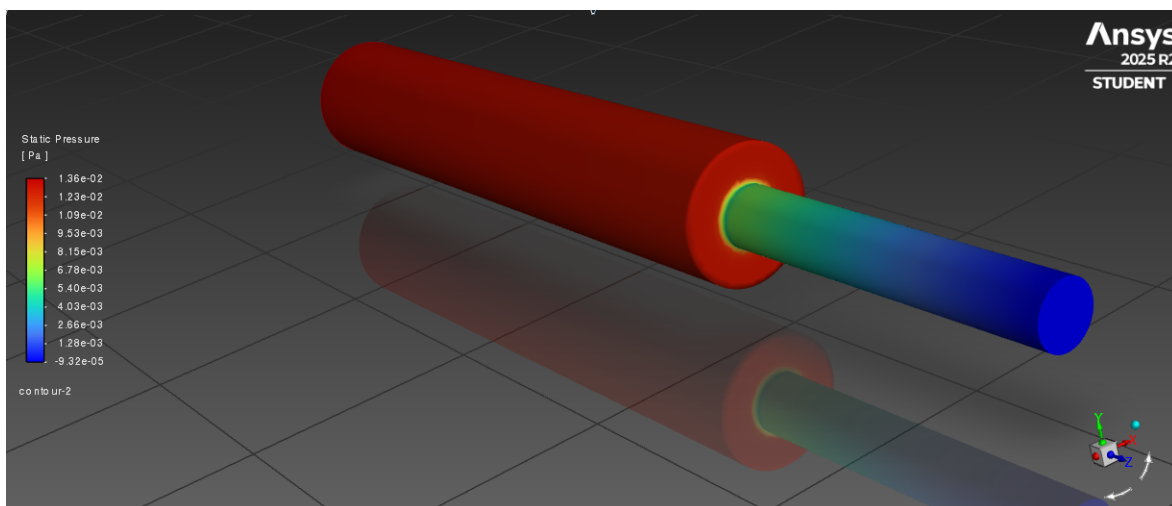


Figure 5: Pressure Contour with Respect to the First Mesh

By inspection, the pressure drop is off by two orders of magnitude relative to the theoretical prediction. This indicates one of the models is faulty. The preliminary calculations ought to be re-examined in the future.

In order to achieve the velocity profile at the one inch offset point from the contraction location as prescribed, I first constructed a plane parallel whose normal pointed in the z-direction, and was positioned $1'' = 25.4$ mm from the origin, which was conveniently placed at the contraction location. Then, I simply constructed a line that spanned the (smaller) cylinder's diameter, but also lived inside that auxiliary plane. The fluid speed profile on that line segment is displayed below:

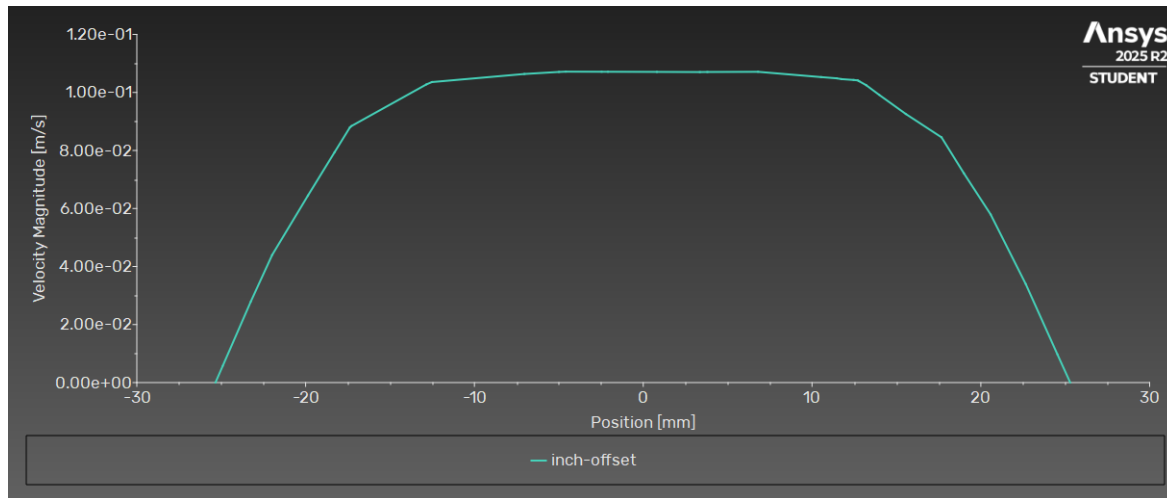


Figure 6: Speed Profile with Respect to the First Mesh

Note that the speed profiles will all be plotted together on a separate distinct plot. This will enable the direct comparison of the fluid velocity after the contraction later on.

The Second Mesh

The second mesh demanded a more structured grid whilst keeping the same cell size (resolution). This was accomplished using multizone and the inflation tool (selecting the cylinder walls), which is illustrated below:

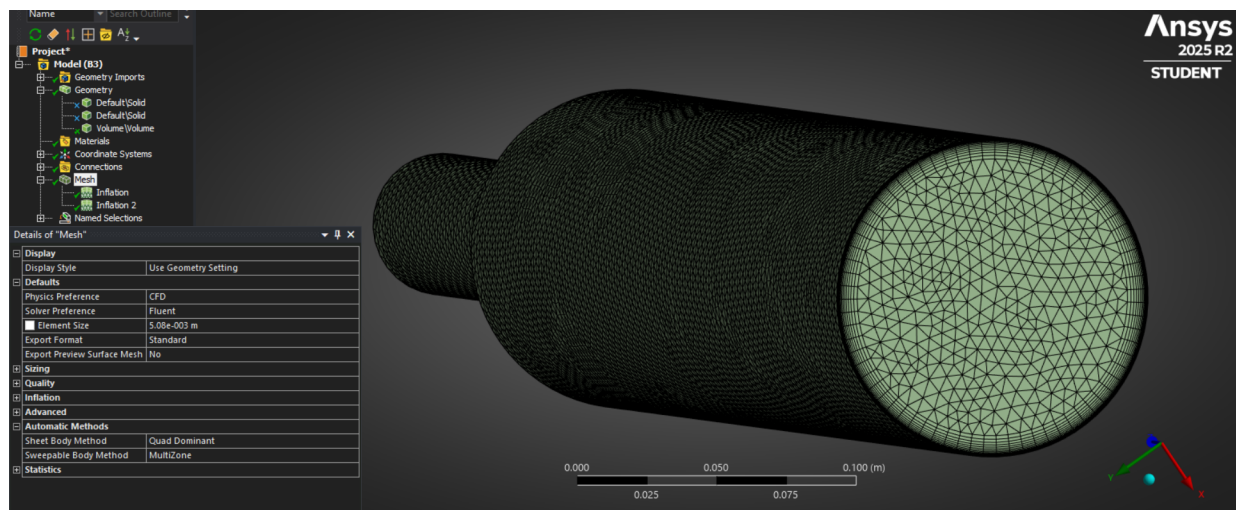


Figure 7: The Second Mesh

This mesh was implemented the exact same way into Fluent. Same boundary conditions, fluid, and physics engine (algorithm). The respective residuals are shown below:

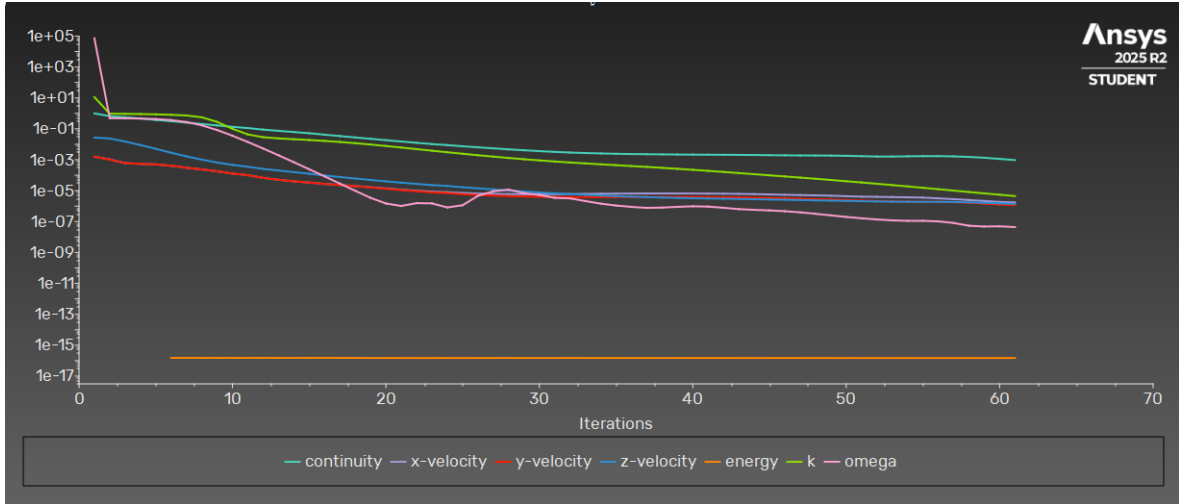


Figure 8: Residuals with Respect to the Second Mesh

The fluid stream lines, pressure contour, and respective speed profile performed similarly (quantitatively) relative to the first mesh, and are illustrated below:

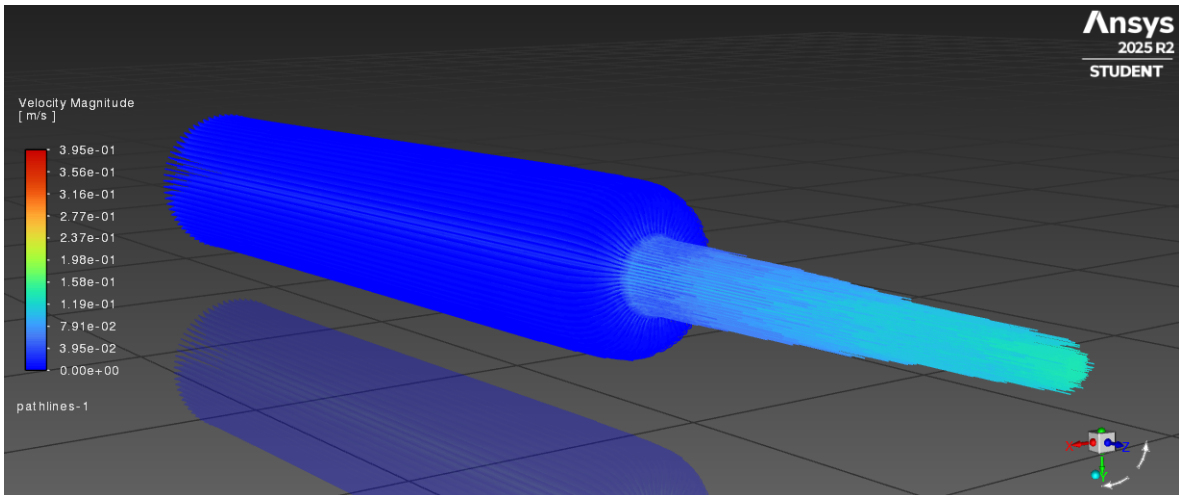


Figure 9: Stream Lines with Respect to the Second Mesh

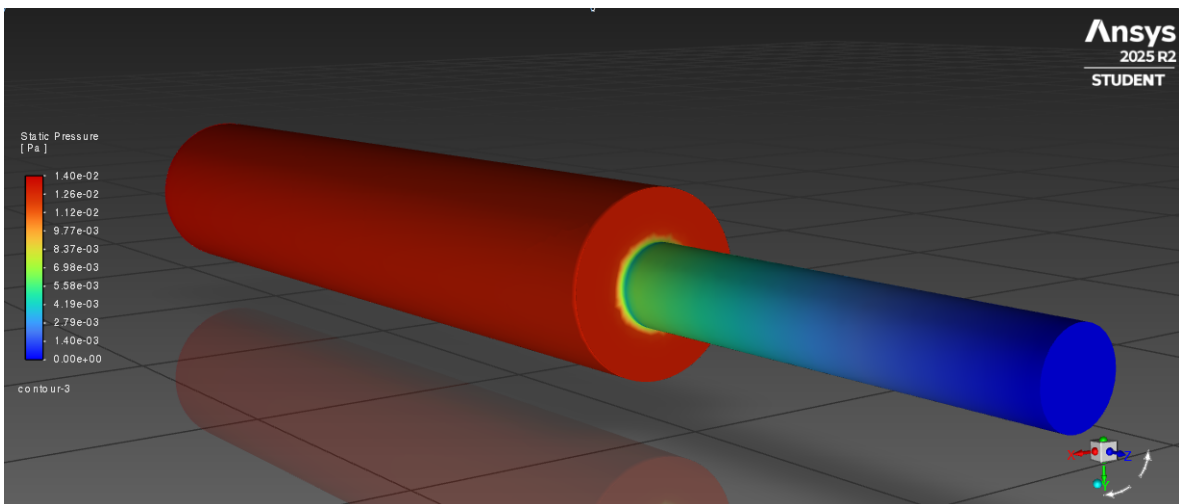


Figure 10: Pressure Contour with Respect to the Second Mesh

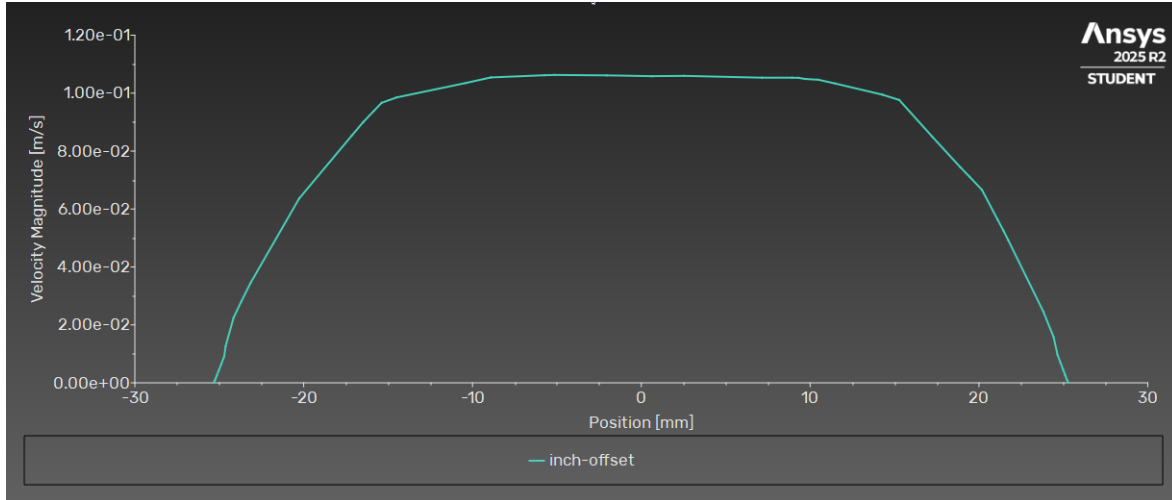


Figure 11: Speed Profile with Respect to the Second Mesh

The Third Mesh

The third mesh will have the same structure as the second, except its cell sizes will be scaled down by a factor of 1.5 in order to increase the resolution. The new cell size was constructed to be $0.13'' = 0.0033$ m.

The reason as to why it was not scaled down by a factor of 2 was because too many cells were generated, and the software restricted me from going beyond a certain threshold.

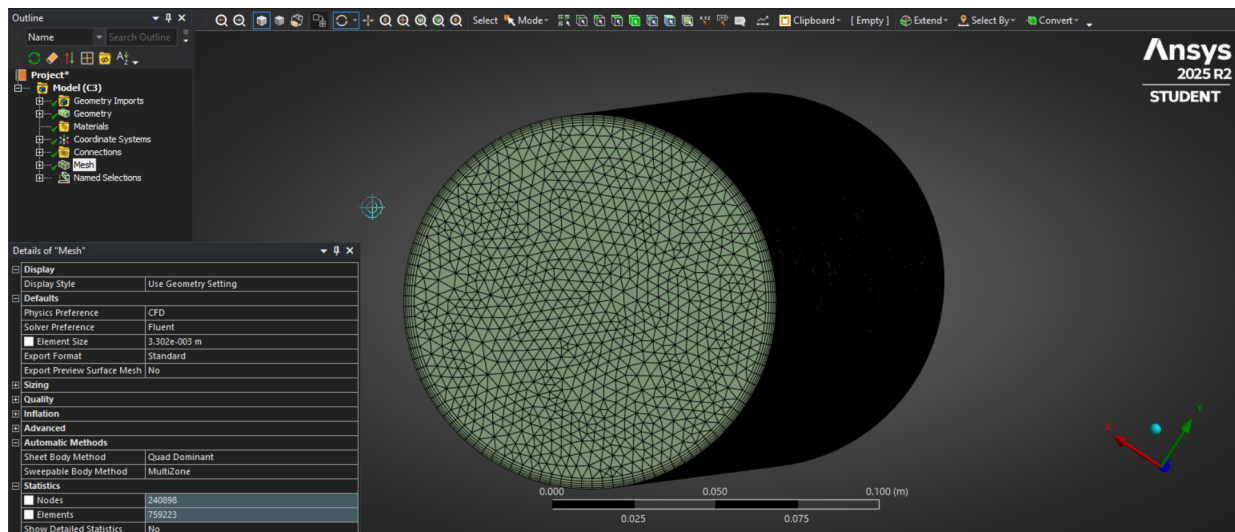


Figure 12: The Third Mesh

Once again, this mesh was implemented the exact same way into Fluent. Same boundary conditions, fluid, and physics engine (algorithm). The respective residuals are shown below:

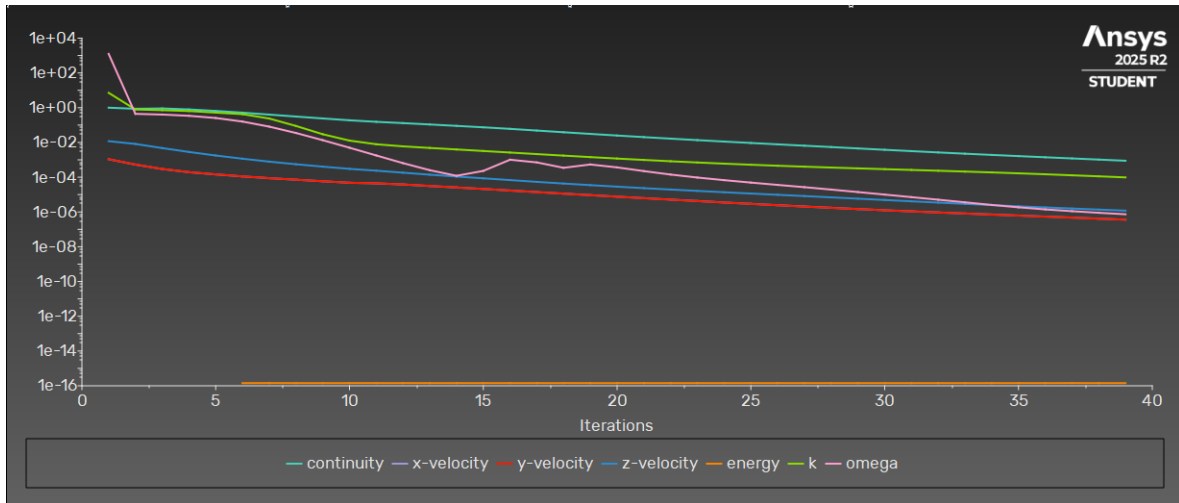


Figure 13: Residuals with Respect to the Third Mesh

The fluid stream lines, pressure contour, and respective speed profile performed similarly (quantitatively) relative to the first mesh, and are once again illustrated below:

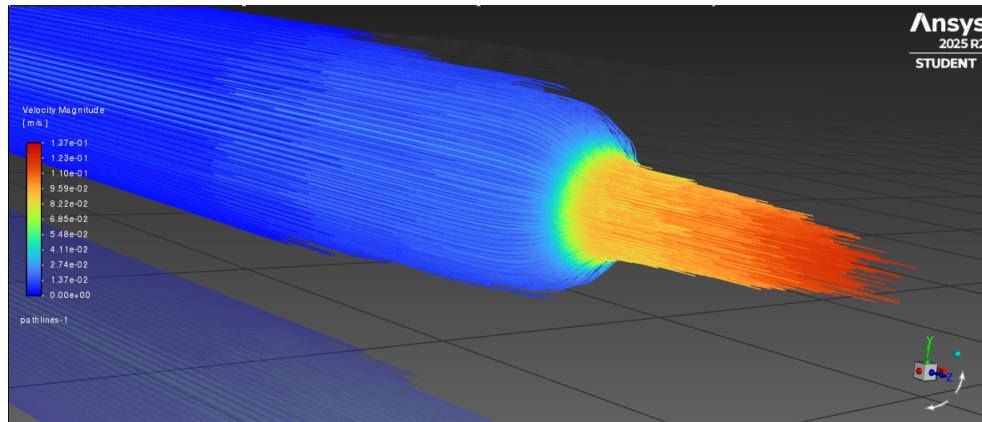


Figure 14: Stream Lines with Respect to the Third Mesh

Just as a side note, I have to say this looks pretty cool. Also, I should have mentioned in the previous stream line visualization that there seemed to be a problematic location (not visible in the image) in which the speed was overestimated (potentially), which threw off the color map. Here (and in the first simulation), that's not the case.

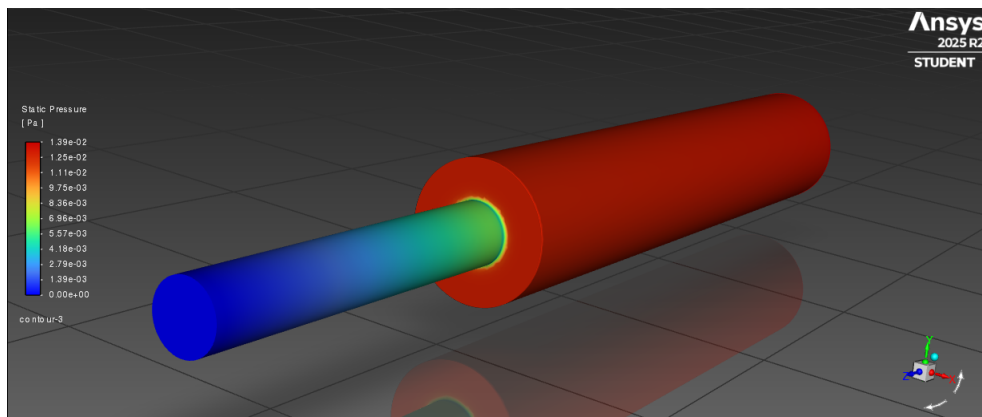


Figure 15: Pressure Contour with Respect to the Third Mesh

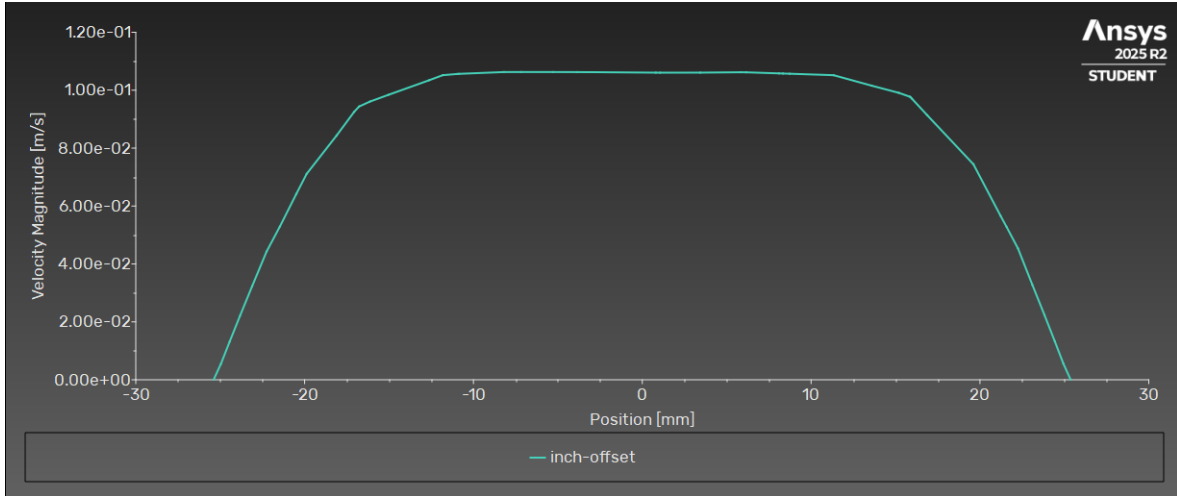


Figure 16: Speed Profile with Respect to the Third Mesh

Comparison

The three solutions from the simulations performed on the three meshes were all forwarded into a single separate CFD-post results section in the workbench. There, the contraction plane and profile line (diameter) were construction as done beforehand. The y-axis was populated by the fluid velocities at each position on the line (x-axis) for each respective simulation.

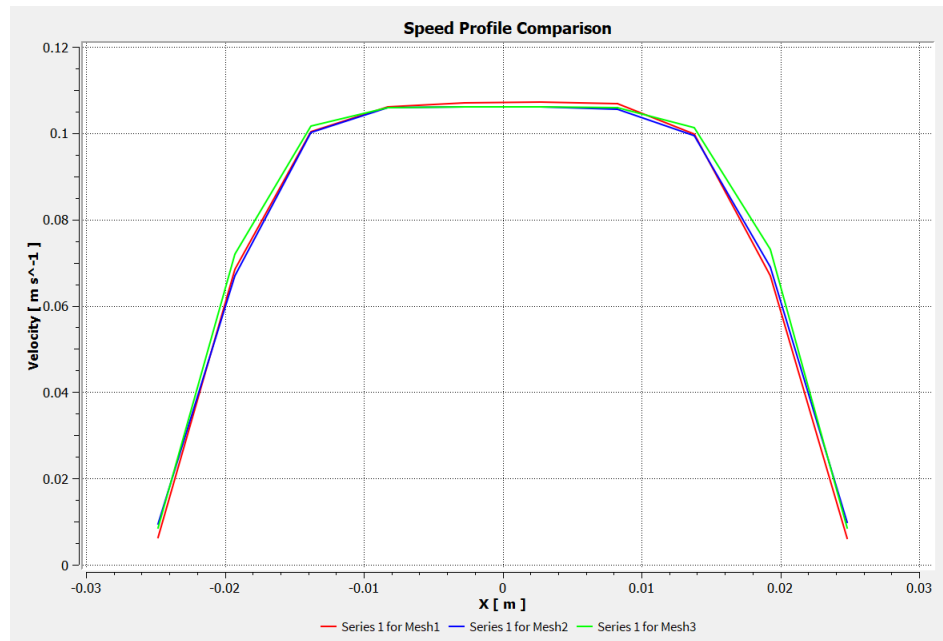


Figure 17: Speed Profile Comparison

One could see that the speed profiles are essentially identical in each case. This loosely implies the three simulations performed similarly at that location. In conjunction with the other contours, a picture is painted that depicts all three simulation behaving similarly (within orders of magnitude), which provides evidence for the legitimacy of this overall modeling procedure.

With respect to the preliminary calculations, the speed predictions all line up quite well, however, the pressure drop was consistently off by two orders of magnitude. Not much can be said other than the preliminary calculations are likely really wrong or the CFD analysis was initialized incorrectly.